STA 4210 HW 2

Yansheng Luo

1. An electrical contractor fits a simple linear regression model, relating cost to wire a house (Y, in dollars) to the size of the house (X, in ft^2). She fits a model, based on a sample of $n = 16$ houses and obtains the following results

$\hat{Y}_i = 50 + 0.22X_i$, $s^2 = 1600$, $\sum_{i=1}^{n}(X_i - \bar{X})^2 = 4000000$, $\bar{X} = 2000$

Given:

$$\hat{Y}_i = 50 + 0.22X_i, \qquad s^2 = 1600, \qquad \sum_{i=1}^{n}(X_i - \bar{X})^2 = 4{,}000{,}000, \qquad \bar{X} = 2000, \qquad n = 16$$

Let $s = \sqrt{s^2} = 40$ and $df = n - 2 = 14$.

a. Compute the standard error of b_1

Solution:

From the formula sheet,

$$S(b_1) = \frac{s}{\sqrt{\sum(X_i - \bar{X})^2}}.$$

```
b1 <- 0.22
s2 <- 1600
s <- sqrt(s2)
Sxx <- 4e+06  #sum of squares

Se_b1 <- s/sqrt(Sxx)  #by formula
Se_b1
```

```
## [1] 0.02
```

$$S(b_1) = \frac{40}{\sqrt{4{,}000{,}000}} = 0.02$$

b. Compute a 95% Confidence Interval for β_1

Solution:

From the formula sheet,

$$b_1 \pm t_{1-\alpha/2,\, n-2}\, S(b_1).$$

```r
df <- 16 - 2  #degrees of freedom
df
```

```
## [1] 14
```

```r
tcrit <- 2.145  #t at 95% confidence, because CI is 2 sided, we look for t_0.975

CI_b1 <- c(b1 - tcrit * Se_b1, b1 + tcrit * Se_b1)
CI_b1
```

```
## [1] 0.1771 0.2629
```

By the CI formula:

$$CI = b_1 \pm t_{1-\alpha/2,\, n-2}\, S_{b_1}$$

With $t_{0.975,14} = 2.145$,

$$0.22 \pm 2.145(0.02) = (0.1771,\ 0.2629).$$

c. Compute a 95% Confidence Interval for σ^2

Solution:

Using the chi-square interval,

$$\left(\frac{(n-2)s^2}{\chi^2_{1-\alpha/2,\, n-2}}, \frac{(n-2)s^2}{\chi^2_{\alpha/2,\, n-2}}\right).$$

```r
alpha <- 0.05
df <- 14

chi_upper <- qchisq(1 - alpha/2, df)
chi_lower <- qchisq(alpha/2, df)

CI_sig2 <- c(df * s2/chi_upper, df * s2/chi_lower)
CI_sig2
```

```
## [1]  857.6149 3979.5861
```

With $\chi^2_{0.975,14} = 26.1189$ and $\chi^2_{0.025,14} = 5.6287$,

$$\left(\frac{14(1600)}{26.1189}, \frac{14(1600)}{5.6287}\right) = (857.6,\ 3979.6).$$

d. Compute a 95% Confidence Interval for the mean of all homes with $X_0 = 2000$

Solution:

$$\hat{Y}_0 = 50 + 0.22(2000) = 490.$$

From the formula sheet,

$$S(\hat{Y}_0) = s\sqrt{\frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum(X_i - \bar{X})^2}}.$$

```r
b0 <- 50  #just assigning the variables again
X0 <- 2000
xbar <- 2000
n <- 16

Yhat0 <- b0 + b1 * X0
Se_Yhat0 <- s * sqrt(1/n + (X0 - xbar)^2/Sxx)

CI_mean <- c(Yhat0 - tcrit * Se_Yhat0, Yhat0 + tcrit * Se_Yhat0)  # compute CI
Yhat0
```

```
## [1] 490
```

```r
Se_Yhat0
```

```
## [1] 10
```

```r
CI_mean
```

```
## [1] 468.55 511.45
```

Since $X_0 = \bar{X}$,

$$S(\hat{Y}_0) = 40\sqrt{\frac{1}{16}} = 10.$$

$$CI = \hat{Y} \pm t_{1-\alpha/2,\, n-2}\, S_{\hat{Y}}$$

$$490 \pm 2.145(10) = (468.55,\ 511.45).$$

e. Compute a 95% Prediction Interval for her brother-in-laws house with $X_0 = 2000$

Solution:

Since we are predicting a value: From the formula sheet,

$$S_{\text{pred}} = s\sqrt{1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{\sum(X_i - \bar{X})^2}}.$$

```r
Se_pred <- s * sqrt(1 + 1/n + (X0 - xbar)^2/Sxx)

PI <- c(Yhat0 - tcrit * Se_pred, Yhat0 + tcrit * Se_pred)
Se_pred
```

```
## [1] 41.23106
```

```
PI
```

```
## [1] 401.5594 578.4406
```

Since $X_0 = \bar{X}$,

$$S_{\text{pred}} = 40\sqrt{1 + \frac{1}{16}} = 41.23.$$

$$PI = \hat{Y} \pm t_{1-\alpha/2,\, n-2}\, S_{\text{pred}}$$

$$490 \pm 2.145(41.23) = (401.56,\ 578.44).$$

2.(R) A substance used in biological and medical research is shipped by airfreight to users in cartons of 1,000 ampules. The data below, involving 10 shipments, were collected on the number of times the carton was transferred from one aircraft to another over the shipment route (X) and the number of ampules found to be broken upon arrival (Y). Assume that the simple linear regression model is appropriate.

```
shipment <- c(1, 0, 2, 0, 3, 1, 0, 1, 2, 0)
ampules <- c(16, 9, 17, 12, 22, 13, 8, 15, 19, 11)
```

Do not use the `lm` and `anova` function for this problem.

a. Compute the ANOVA table. Please print out the SS and MS for each source and the corresponding degree of freedom.

Solution:

```
options(digits = 6)  # 6 sig fig
x <- shipment
y <- ampules
n <- length(x)

## compute by formula given from the lecture
xbar <- mean(x)
ybar <- mean(y)

Sxx <- sum((x - xbar)^2)
Sxy <- sum((x - xbar) * (y - ybar))

b1 <- Sxy/Sxx
b0 <- ybar - b1 * xbar

yhat <- b0 + b1 * x
SST <- sum((y - ybar)^2)
SSR <- sum((yhat - ybar)^2)
SSE <- sum((y - yhat)^2)

MSR <- SSR/1
MSE <- SSE/(n - 2)

n
```

```
## [1] 10
```

```r
n - 1
```

```
## [1] 9
```

```r
n - 2
```

```
## [1] 8
```

```r
cat(sprintf("SST = %.1f\n", SST))
```

```
## SST = 177.6
```

```r
cat(sprintf("SSR = %.1f\n", SSR))
```

```
## SSR = 160.0
```

```r
cat(sprintf("SSE = %.1f\n", SSE))
```

```
## SSE = 17.6
```

```r
cat(sprintf("MSR = %.1f\n", MSR))
```

```
## MSR = 160.0
```

```r
cat(sprintf("MSE = %.2f\n", MSE))
```

```
## MSE = 2.20
```

The resulting ANOVA table is:

Source	SS	df	MS
Regression	160.0	1	160.0
Error	17.6	8	2.20
Total	177.6	9	

b. Conduct the F-test of $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 \neq 0$ with $\alpha = 0.05$. Report the p-value and state your conclusion.

Solution:

```r
Fstat <- MSR/MSE  #f-calculated
pval <- 1 - pf(Fstat, 1, 8)

Fstat
```

```
## [1] 72.7273
```

```
pval
```

```
## [1] 2.74867e-05
```

We test:

$$H_0: \ \beta_1 = 0 \quad \text{vs.} \quad H_1: \ \beta_1 \neq 0.$$

The test statistic is

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{160.0}{2.20} = 72.73.$$

Under H_0,

$$F \sim F_{1,8}.$$

The corresponding p-value (from R) is extremely small:

$$p < 0.0001.$$

Conclusion:
Since the p-value is much smaller than $\alpha = 0.05$, we reject H_0. There is strong statistical evidence that the number of damaged ampules is linearly associated with the number of aircraft transfers during shipment.

3.(R) A criminologist collected data on the percentage of individuals having at least a high-school diploma (X) and the crime rate (Y, number of crimes per 100,000 residents) in 84 medium-sized US counties. You can use any R functions for this problem.

```
crime <- read.table("http://www.stat.ufl.edu/~rrandles/sta4210/Rclassnotes/data/textdatasets/KutnerData/
    col.names = c("y", "x"))
```

a. Fit the linear regression model using `lm()`.

Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

```
fit <- lm(y ~ x, data = crime)

summary_fit <- summary(fit)
summary_fit
```

```
##
## Call:
## lm(formula = y ~ x, data = crime)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -5278  -1758   -210   1575   6803
##
## Coefficients:
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20517.6     3277.6    6.26  1.7e-08 ***
## x             -170.6       41.6   -4.10  9.6e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2360 on 82 degrees of freedom
## Multiple R-squared:  0.17,   Adjusted R-squared:  0.16
## F-statistic: 16.8 on 1 and 82 DF,  p-value: 9.57e-05
```

Fitted equation:

$$\hat{Y} = 20517.6 - 170.6X$$

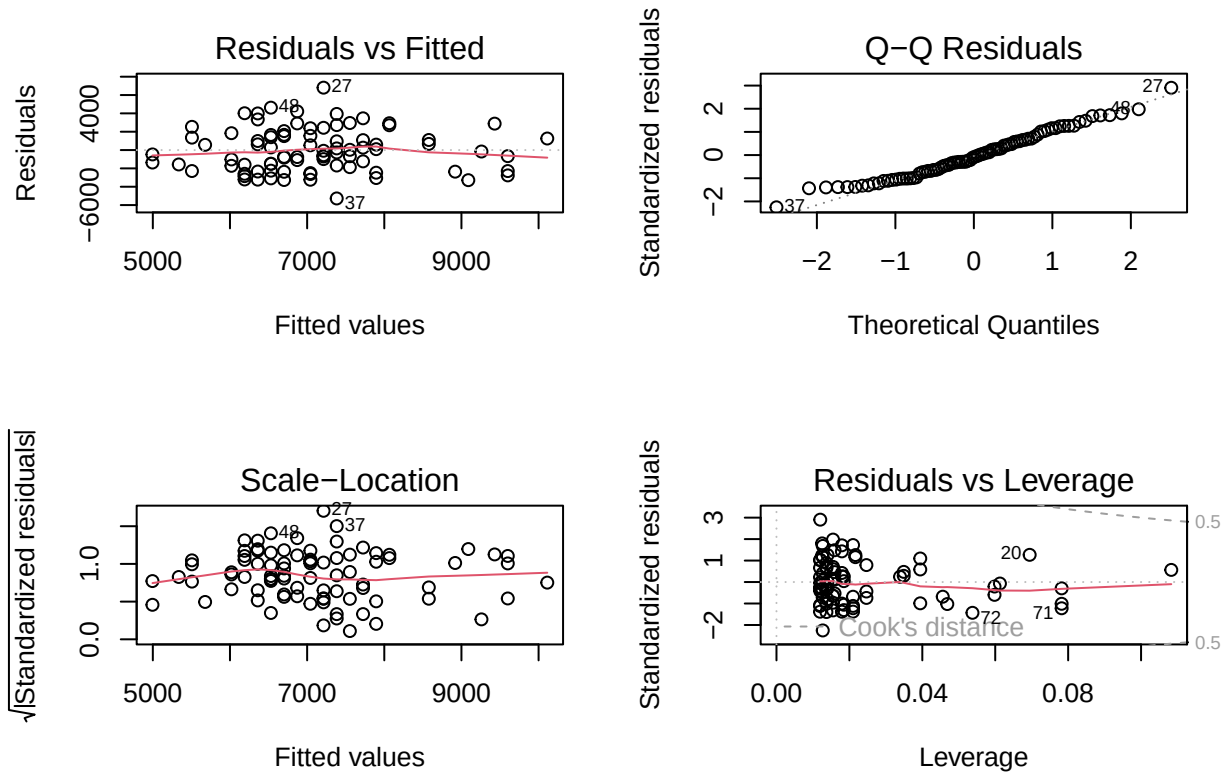
The slope estimate is negative, indicating that counties with a higher percentage of individuals holding at least a high-school diploma tend to have lower crime rates. Specifically, for each 1% increase in the high-school graduation rate, the crime rate is estimated to decrease by approximately 170.6 crimes per 100,000 residents.

The slope is statistically significant ($p = 9.6 \times 10^{-5}$), indicating strong evidence of a linear association between education level and crime rate. The model explains approximately 17% of the variability in crime rates ($R^2 = 0.17$), suggesting that while education is important, other factors also contribute to crime levels.

b. Generate plots to check the four linear regression assumptions, and explain whether the assumptions are violated or not.

Assumptions: 1. Linearity
2. Normality
3. Independence
4. Constant variance

```r
par(mfrow = c(2, 2))
plot(fit)
```




```
par(mfrow = c(1, 1))
```

Interpretation:

Linearity:
The Residuals vs Fitted plot does not show strong curvature or a pattern, which indicates that a linear relationship between X and Y is reasonable.

Normality:
The Normal Q-Q plot shows the residuals lying approximately along a straight line.It suggests that the normality assumption is reasonable.

Constant Variance:
The Scale-Location plot does not display a clear funnel shape. The spread of residuals appears relatively constant across fitted values, supporting the homoscedasticity assumption.

Influential Points:
The Residuals vs Leverage plot does not show extreme influential observations based on Cook's distance. Points with moderate leverage (like the one near 0.08) have small residuals, so they are not distorting the fitted line.

c. Perform the following statistical tests with $\alpha = 0.05$ to further evaluate the model assumptions:

- Shapiro-Wilk test
- Runs test
- Durbin-Watson test

- Levene's test
- Breusch-Pagan/Cook-Weisberg test
- Lack-of-fit test
  Interpret the results of each test. If you think any of these tests are not applicable in this case, explain your reasoning.

–

(1) Shapiro–Wilk test (Normality)

$$H_0 : \text{Errors are normally distributed}$$

```r
# install dependencies
if (!require(randtests)) install.packages("randtests")
```

```
## Loading required package: randtests
```

```r
if (!require(lmtest)) install.packages("lmtest")
```

```
## Loading required package: lmtest

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
if (!require(car)) install.packages("car")
```

```
## Loading required package: car

## Loading required package: carData
```

```r
library(randtests)
library(lmtest)
library(car)

resid_vec <- residuals(fit)
shapiro_res <- shapiro.test(resid_vec)
shapiro_res
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resid_vec
## W = 0.9776, p-value = 0.151
```

The p-value is 0.151. Since $p > 0.05$, we fail to reject H_0. There is no significant evidence that the residuals deviate from normality.

---

(2) Runs test (Independence)

$$H_0 : \text{Errors are independent}$$

```r
library(randtests)

run_res <- runs.test(resid_vec)
run_res
```

```
##
##  Runs Test
##
## data:  resid_vec
## statistic = -2.635, runs = 31, n1 = 42, n2 = 42, n = 84, p-value =
## 0.00843
## alternative hypothesis: nonrandomness
```

The p-value is 0.00843. Since $p < 0.05$, we reject H_0. There is evidence of non-randomness in the residuals.

---

(3) Durbin–Watson test (Autocorrelation)

$$H_0 : \rho = 0$$

$$TS = \frac{\sum_{i=2}^{n}(e_i - e_{i-1})^2}{\sum_{i=1}^{n} e_i^2}$$

```r
library(lmtest)

dw_res <- dwtest(fit)
dw_res
```

```
##
##  Durbin-Watson test
##
## data:  fit
## DW = 1.495, p-value = 0.0087
## alternative hypothesis: true autocorrelation is greater than 0
```

The p-value is 0.0087. Since $p < 0.05$, we reject H_0. There is evidence of positive autocorrelation, indicating that the independence assumption may be violated.

---

(4) Levene's test (Homogeneity of variance) Artificial grouping since X is quantitative.

$$H_0 : \sigma_1^2 = \sigma_2^2 = \cdots$$

```r
library(car)

crime$group <- cut(crime$x, breaks = 4)

levene_res <- leveneTest(resid_vec ~ group, data = crime)
levene_res
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  3   0.685  0.564
##       80
```

The p-value is 0.564. Since $p > 0.05$, we fail to reject H_0. There is no evidence of unequal variances across groups. Note that grouping was artificial since X is quantitative.

---

(5) Breusch–Pagan test (Homoscedasticity)

$$H_0 : \gamma_1 = 0$$

```r
bp_res <- bptest(fit)
bp_res
```

```
##
##  studentized Breusch-Pagan test
##
## data:  fit
## BP = 0.006095, df = 1, p-value = 0.938
```

The p-value is 0.938. Since $p > 0.05$, we fail to reject H_0. There is strong evidence that the variance is constant.

---

(6) Lack-of-fit test Requires repeated X levels.

```r
length(unique(crime$x))
```

```
## [1] 24
```

A formal lack-of-fit test requires replicated X values to estimate pure error. Since most X values are distinct, a classical lack-of-fit test is not appropriate in this setting.

$$df_{pure\ error} = n - t = 0$$

Therefore, lack-of-fit test is not applicable.

Overall Conclusion

The linearity, normality, and constant variance assumptions appear reasonable based on both graphical diagnostics and formal tests. However, both the runs test and Durbin–Watson test indicate evidence of dependence (autocorrelation) in the residuals.

Thus, while the regression model suggests a statistically significant negative relationship between education level and crime rate, caution should be exercised because the independence assumption may be violated.